

DecisionOS — Product Documentation (Start to End)

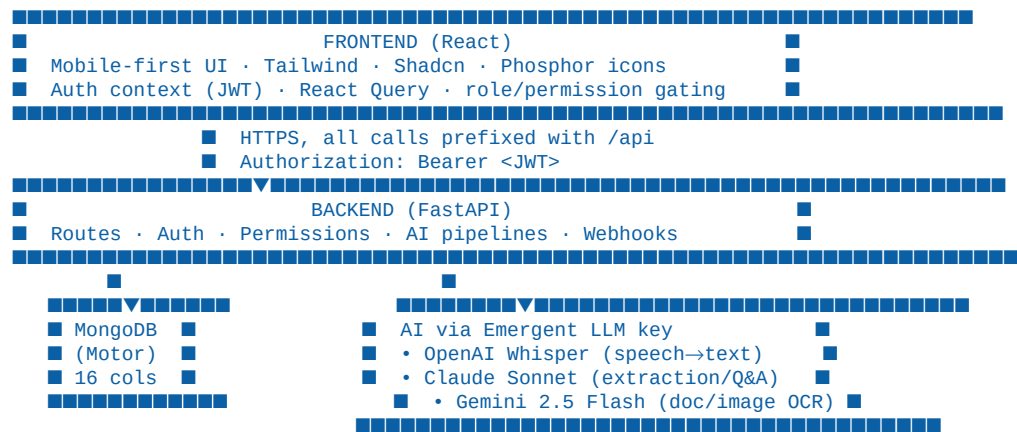
The AI operating brain for founder-led SMEs — capture decisions, turn them into structured work, and give the team one shared source of truth.

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1. What DecisionOS Does (in one line)

A founder speaks or forwards something (a voice note, a typed directive, an invoice PDF, an Excel sheet, or later a WhatsApp message). DecisionOS **understands it with AI**, turns it into **structured records** (decisions, tasks, invoices, payments, contacts, complaints, reminders), **assigns the work** to the right person, and surfaces everything in **one Inbox + a daily CEO Brief + a searchable Company Brain**.

2. High-Level Architecture



- **Multi-tenant:** every record carries a `tenant_id`; users only ever see their own workspace.
- **Environment-driven:** URLs, DB, secrets and API keys come from `.env` only.
- **AI is provider-agnostic:** all AI runs through the built-in **Emergent LLM key** (no separate OpenAI/Anthropic/Google keys needed).

3. Tech Stack

Layer & Technology

Frontend & React, React Router, React Query, TailwindCSS, Shadcn UI, Phosphor icons, Sonner (toasts)

Backend & FastAPI, Motor (async MongoDB), Pydantic

Database & MongoDB (16 collections)

Auth & JWT (email + password), bcrypt hashing

AI & Emergent LLM key → OpenAI Whisper, Anthropic Claude Sonnet, Google Gemini 2.5 Flash

Files/Media & Local upload store served at `/api/files/{name}`

Integrations (wired) & WhatsApp Cloud API (Meta) — awaiting keys; Resend (email) — optional key

4. Data Model (MongoDB collections)

Collection & Purpose & Key fields

tenants & The company/workspace & name, industry, company_size, region, currency, gst, branches, business_scale, current_software, roles[], products[], invited_employees[]

users & Team members & tenant_id, name, email, password_hash, role, **permissions**[]

voice_notes & Raw voice/text captures & tenant_id, kind (audio/text), transcript, language, status, source

decisions & AI-structured directives & tenant_id, title, summary, items[], dtype, confidence, status (pending_approval/approved/rejected), task_ids[]

tasks & Actionable work & tenant_id, title, **assignee_id**, **assignee_role**, priority, status (blocked/todo/in_progress/done), due_date, source

workflows & Multi-stage processes & tenant_id, type (sales_dispatch/purchase_payment), stage, contact_id

contacts & CRM: customers & vendors & tenant_id, type, name, company, phone, email, address, tax_id, tags, status

invoices & Sales invoices & purchase bills & tenant_id, type, number, contact_id, amount, currency, status, line_items[]

payments & Money in/out & tenant_id, direction (in/out), amount, method, reference, contact_id

complaints & Customer complaints & tenant_id, customer_id, text, severity, status

inbox & Unified feed items & tenant_id, source, **classification**, title, ref_type, ref_id, status

ingestions & Upload/OCR jobs & tenant_id, source, kind, filename, doc_type, records{}, status

attendance & Present/absent marks & tenant_id, user_id, status, date

notifications & In-app alerts & tenant_id, user_id, message, type, is_read

memory & Company memory notes & tenant_id, text, tag

activity & Audit trail of actions & tenant_id, actor, kind, message, entity

5. Roles & Permissions

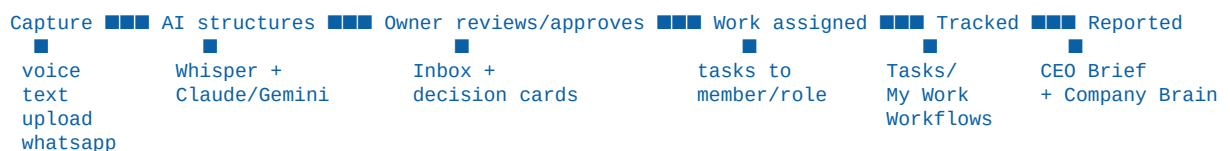
- **Roles** are per-tenant and industry-aware (Owner + AI-suggested roles like Sales, Production, Finance).
- **Owner** always has full access.
- **Module-level permissions** per user: **inbox**, **data_input**, **people**, **finance**, **workflows**, **tasks**, **brain**, **ask**, **team_manage**.
- Users with no explicit permissions fall back to **role defaults**:
- Sales → base + data_input + people
- Finance → base + data_input + finance
- Others → base (inbox, people, workflows, tasks, brain, ask)
- Enforcement is **both** frontend (hidden nav/actions) **and** backend (**require_perm**) for the sensitive areas: Finance (money), Data Input (uploads), Team management.

6. End-to-End User Flow

6.1 Onboarding — the "Digital Executive Office" (7 steps)

1. **About your company** — company name, owner account (name/email/password), industry, GST (optional), branches.
2. **Business scale** — team size, currency, monthly sales/purchases, # customers/suppliers, region.
3. **Current software** — Excel / Tally / Zoho / Others.
4. **AI Team & Products** — once industry is known, AI proposes team roles + products/services; owner edits, then **workspace is created** (authenticated from here on).
5. **Connect business** — upload Excel/CSV now (auto-imported); Tally/Zoho shown as "coming soon".
6. **Invite employees** — by mobile number (stored as pending invites; real SMS once Twilio connected).
7. **AI learns business** — animated setup (Company Brain, import, index, dashboard) → **Enter DecisionOS**.

6.2 Daily working loop



Step-by-step:

1. **Capture** — Owner taps the mic or types in the **Inbox**, or uploads a PDF/CSV in **Data Input**, or (when live) forwards a document on **WhatsApp**.
2. **AI understanding** — Whisper transcribes audio; Claude extracts decisions/tasks/reminders/memory; Gemini reads invoices/images. A **specific team member named** in the directive is auto-assigned; otherwise the task is auto-assigned **by role**.
3. **Unified Inbox** — every item appears in one feed, **auto-classified** as Customer / Supplier / Invoice / Payment / Complaint / Task / Approval / Reminder. Owner can mark done/dismiss.
4. **Approve** — decisions start as *pending approval*; tasks are *blocked* until the owner approves (then they become actionable).
5. **Execute** — team members see their work in **My Work** and the **Tasks** kanban (To Do → In Progress → Done); anyone can reassign.
6. **Track processes** — **Workflows** handle multi-stage flows (sales dispatch, purchase→payment) with stage advancement.
7. **Report** — the **CEO Brief** shows the box-style dashboard (delayed, completed, awaiting approval, absent, complaints, overdue payments); the **Company Brain** lets you search everything and **Ask AI** natural-language questions.

7. Screens (Frontend Pages)

Page	Route	What it does
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Login / Onboarding /login Sign in + 7-step registration wizard

Inbox (home) / Unified capture + AI-classified feed + pending approvals

Daily Brief (Dashboard) [/dashboard](#) Operational overview counters

CEO Brief | /[brief](#) | Box-style KPI dashboard (Morning/Evening/Weekly/Monthly)
Data Input | /[ingest](#) | Upload PDF/image/CSV → review → file; view invoices/payments
Workflows | /[workflows](#) | Multi-stage process kanban
People | /[contacts](#) | CRM list of customers/vendors
360° Profile | /[contacts/:id](#) | Full customer/supplier view (owner+finance)
Tasks | /[tasks](#) | Kanban; assign/reassign to members
My Work | /[my-work](#) | An employee's own tasks
Company Brain | /[brain](#) | Keyword search across all records
Ask AI | /[ask](#) | Natural-language Q&A with citations
Team | /[team](#) | Members, roles, per-user access control, invites
Notifications | /[notifications](#) | In-app alerts

8. AI Features (the intelligent core)

8.1 Voice & Text → Structured work

- **Whisper** transcribes voice notes in **English, Tamil, and Tanglish** (code-mixed).
- **Claude Sonnet** converts the transcript into a **decision + tasks + reminders + memory notes**, all output in clean English.
- **Named-person assignment**: if the directive says "tell Priya...", the task is assigned to that member; otherwise it's assigned by **role** inferred from the work.

8.2 Document & Data Ingestion (OCR)

- **Gemini 2.5 Flash (vision)** reads invoice/bill/payment **PDFs and images**, extracting parties, amounts, dates, line items — and auto-creates follow-up tasks.
- **CSV/Excel auto-detect**: Claude classifies a spreadsheet (customers/vendors/invoices/payments) and maps every row — no manual column mapping.
- All ingestion flows through one pipeline → editable **review** → **file** into the database.

8.3 Unified Inbox auto-classification

- Every incoming item (voice, upload, complaint, WhatsApp) is tagged into one of 8 categories automatically, so the owner has a single screen to work from.

8.4 Company Brain — Search & Ask AI

- **Search** tokenizes your query and matches across decisions, tasks, workflows, contacts, invoices and memory.
- **Ask AI** answers natural-language questions grounded ONLY in your company data, returning **citations**. For Owner/Finance it also answers money questions ("who owes more than **₹5 lakh**?", "which customer hasn't paid in 30 days?", "yesterday's sales") using invoices, payments, per-party outstanding and today's date.

8.5 Industry-aware onboarding

- Given the industry, AI proposes suitable **team roles** and **products/services** to pre-populate the workspace.

8.6 CEO Brief & Follow-up engine

- Aggregates delayed/completed/awaiting-approval tasks, absent employees, open complaints and overdue payments; a follow-up/escalation engine raises in-app notifications.

9. Key API Endpoints (grouped)

- **Auth:** POST /api/auth/register, POST /api/auth/login, GET /api/auth/me
- **Onboarding:** POST /api/onboarding/suggest, GET/POST /api/invites
- **Capture & AI:** POST /api/voice-notes (audio), POST /api/voice-notes/text, GET /api/voice-notes, POST /api/ask, GET /api/brain/search
- **Decisions:** GET /api/decisions, POST /api/decisions/{id}/approve|reject
- **Tasks:** GET/POST /api/tasks, PATCH /api/tasks/{id}, POST /api/tasks/{id}/attachment
- **Ingestion:** POST /api/ingest/document, POST /api/ingest/csv, POST /api/ingest/{id}/commit, GET /api/ingest
- **Money:** GET /api/invoices, GET /api/payments, GET /api/contacts/{id}/profile
- **CRM & complaints:** GET/POST /api/contacts, PATCH/DELETE /api/contacts/{id}, GET/POST /api/complaints
- **Inbox:** GET /api/inbox, POST /api/inbox/{id}/status
- **Workflows:** GET/POST /api/workflows, PATCH /api/workflows/{id}/advance
- **Team:** GET/POST /api/users, PATCH /api/users/{id}, GET/POST /api/attendance
- **Dashboards:** GET /api/dashboard, GET /api/brief, POST /api/brief/send-digest
- **Notifications:** GET /api/notifications, POST /api/notifications/{id}/read, POST /api/follow-up/run
- **WhatsApp:** GET/POST /api/webhooks/whatsapp

10. Integrations Status

Integration Status Notes

AI (Whisper, Claude, Gemini) ■ Live via Emergent LLM key

WhatsApp Cloud API (Meta) ■ Wired, awaiting keys webhook verify + media OCR + auto-file + reply implemented

Resend (email digest) ■ Optional mocked/logged until key added

Twilio (SMS invites) ■ Planned invites stored as pending

Zoho Books ■ Planned needs Client ID/Secret

[illegible]

11. Security & Multi-tenancy

- JWT bearer auth on every `/api` call; passwords bcrypt-hashed.
- Every query is scoped by `tenant_id` — no cross-company data leakage.
- Sensitive modules (money, uploads, team management) enforced server-side via permissions.
- WhatsApp webhook validates Meta's `X-Hub-Signature-256` when the app secret is set.

12. Glossary

- **Decision** — an AI-structured directive from the owner, awaiting approval.
- **Task** — an actionable item, assigned to a member or a role, tracked on the kanban.
- **Workflow** — a multi-stage business process (e.g., dispatch, purchase→payment).
- **Ingestion** — an uploaded/forwarded document processed by AI into records.
- **Company Brain** — the searchable + askable knowledge layer over all your data.